

LLMs for Antibiotic Prescription and Antimicrobial Resistance

Overview

This project is a literature review that explores how Large Language Models (LLMs) such as ChatGPT are being used in antibiotic prescribing, antimicrobial stewardship, and antimicrobial resistance (AMR) research.

The goal of the project was to understand the current state of research in this area and evaluate whether these technologies are ready to be used in real healthcare settings. The review also examines policy and governance considerations based on international AI guidelines.

Objectives

- Review recent research on LLMs in antibiotic prescribing and AMR.
- Compare findings across multiple studies.
- Identify common strengths and limitations of current models.
- Analyze the results from a healthcare policy and governance perspective.

Dataset

A structured spreadsheet was created containing information extracted from selected research papers, including:

- Study details
- Methodology
- Models evaluated
- Key findings
- Limitations
- Future research directions
- Policy and ethical considerations

Key Findings

- LLMs show promise as clinical decision-support tools.
- Performance varies significantly across different models and tasks.
- Accuracy decreases in complex clinical scenarios.
- Human oversight remains necessary for safe healthcare use.
- More validation and governance measures are needed before widespread deployment.

Repository Contents

- `literature_review_report.pdf` – Final project report

- `extracted_studies.xlsx` – Structured dataset of reviewed papers
- `methodology.md` – Summary of the review process

Future Work

Possible extensions include reviewing newer LLMs, expanding the dataset, and exploring real-world clinical validation studies.